SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. A traveling wave is of the form $y(x,t) = A\cos(kx - \omega t) + B\sin(kx - \omega t)$, which can also be written as $y(x,t) = D\sin(kx - \omega t - \phi)$ where

(A) $D = A + B$ (B) $D = |A + B|$ (C) $D^2 = A^2 + B^2$ (D) $D = A - B$

2. A thin string with linear density μ is joined to a thick string with linear density 2μ. A incident pulse is sent down the thin string toward the thick string and eventually creates reflected and transmitted pulses. Which of the following is true ?

(A) The reflected and transmitted pulses are both inverted.

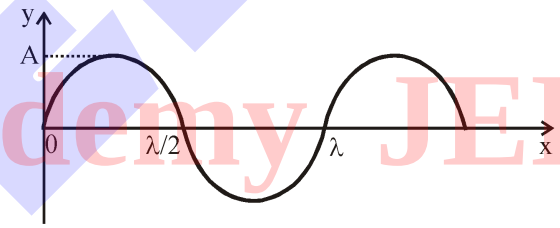
(B) Neither the reflected nor transmitted pulses are inverted.

(C) The reflected pulse is inverted, but the transmitted pulse is not inverted.

(D) The transmitted pulse is inverted, but the reflected pulse is not inverted.

3. Here given snap shot of a progressive wave at $t = 0$ with time period = T. Then the equation of the wave if wave is going in +ve x-direction and if wave is going in –ve x-direction will be respectively.

$\left(\text{Here, } T = \frac{2\pi}{\omega}\right)$



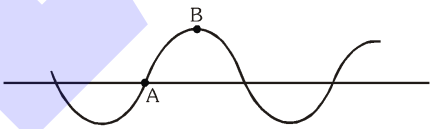
(A) $y = A\sin(kx + \omega t)$, $y = A\sin(kx - \omega t)$

(B) $y = A\cos(kx + \omega t)$, $y = A\cos(kx - \omega t)$

(C) $y = A\sin(\omega t - kx)$, $y = A\sin(\omega t + kx)$

(D) $y = A\sin(kx - \omega t)$, $y = A\sin(kx + \omega t)$

4. A progressive wave is travelling in a string as shown. Then which of the following statement about KE and potential energy of the elements A and B is true?



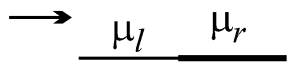
(A) For point A : kinetic energy is maximum and potential energy is min.

(B) For point B : kinetic energy is minimum and potential energy is min.

(C) For point A : kinetic energy is minimum and potential energy is max.

(D) For point B : kinetic energy is minimum and potential energy is max.

5. A string consists of two parts attached at $x = 0$. The right part of the string ($x > 0$) has mass μ_r per unit length and the left part of the string ($x < 0$) has mass μ_l per unit length. The string tension is T. If a wave of unit amplitude travels along the left part of the string, as shown in the figure, what is the amplitude of the wave that is transmitted to the right part of the string ?



(A) 1 (B) $\frac{2}{1+\sqrt{\mu_l/\mu_r}}$ (C) $\frac{2\sqrt{\mu_l/\mu_r}}{1+\sqrt{\mu_l/\mu_r}}$ (D) $\frac{\sqrt{\mu_l/\mu_r}-1}{\sqrt{\mu_l/\mu_r}+1}$

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6. Five waveforms moving with equal speeds on the x-axis

$y_1 = 8\sin(\omega t + kx)$; $y_2 = 6\sin(\omega t + \frac{\pi}{2} + kx)$; $y_3 = 4\sin(\omega t + \pi + kx)$; $y_4 = 2\sin(\omega t + \frac{3\pi}{2} + kx)$;

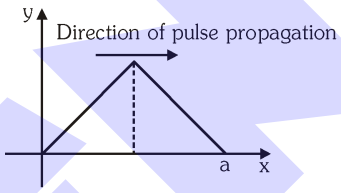
$y_5 = 4\sqrt{2}\sin(\omega t - kx + \frac{\pi}{4})$ are superimposed on each other. The resulting wave is :

(A) $8\sqrt{2}\cos kx \sin(\omega t + \frac{\pi}{4})$ (B) $8\sqrt{2}\sin(\omega t - kx + \frac{\pi}{4})$

(C) $8\sqrt{2}\sin kx \cos(\omega t + \frac{\pi}{4})$ (D) $8\sin(\omega t + kx)$

7. The wave function of a triangular wave pulse is defined by the relation below at time t = 0 sec .

$$y = \begin{cases} mx & for\ 0 \le x \le \frac{a}{2} \\ -m(x-a) & for\ \frac{a}{2} \le x \le a \\ 0 & every\ where\ else, \text{ where } m<<1 \end{cases}$$



The wave pulse is moving in the +X direction in a string having tension T and mass per unit length μ. The total energy present with the wave pulse is :-

(A) $\frac{m^2Ta}{2}$ (B) m^2Ta (C) mTa (D) $\frac{mTa}{2}$

8. A plane progressive harmonic wave is given by the equation : $\varphi = \varphi_m \sin(2t - 3x + 4y + \pi/3)$, where x and y are in meters, and t is in seconds. Let $\hat{n}$ is the unit vector in the direction of wave propagation, and v is the speed of wave w.r.t. the wave medium, then :-

(A) $\hat{n} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$; $v = 0.4$ m/s (B) $\hat{n} = \frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}$; $v = \frac{2}{3}$ m/s

(C) $\hat{n} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$; $v = 0.4$ m/s (D) $\hat{n} = -\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$; $v = 0.5$ m/s

9. The vibrations of a string of length 600 cm fixed at both ends are represented by the equation $y = 4\sin\left(\frac{\pi x}{15}\right)\cos(96\pi t)$, where x and y are in cm and t in second. What is the maximum displacement at a point at x=5 cm?

(A) $\sqrt{3}$ cm (B) $2\sqrt{3}$ cm (C) $3\sqrt{3}$ cm (D) $4\sqrt{3}$ cm

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Multiple Correct Answer Type 9 Q. [4 M (–1)]

10. The figures represent two snaps of a travelling wave on a string of mass per unit length, $\mu = 0.25$ kg/m.

The two snaps are taken at time t = 0 and at $t = \frac{1}{24}$ s. Then the possible solution for wave are :

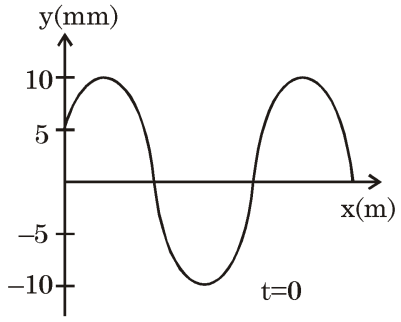


Figure-1

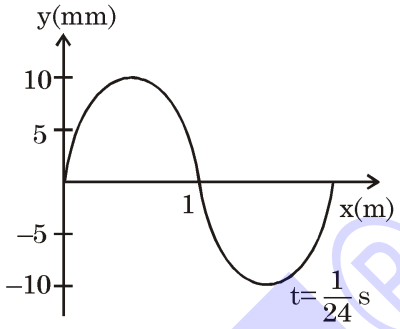


Figure-2

(A) speed of wave is 4 m/s.
(B) the tension in the string is 4 N
(C) the equation of the wave is $y = 10 \sin\left(\pi x - 4\pi t + \frac{\pi}{6}\right)$
(D) the maximum velocity of the particle = $\frac{\pi}{25}$ m/s

11. $y(x, t) = 0.8/[(4x + 5t)^2 + 5]$ represents a moving pulse, where x & y are in meter and t in second. Then:
(A) pulse is moving in +x direction
(B) in 2s it will travel a distance of 2.5 m
(C) its maximum displacement is 0.16 m
(D) it is a symmetric pulse.

12. A string of mass 0.2 kg and length 2m is tied at two ends to fixed supports under a tension of 10 N. A point P on the string is found to travel from one extreme to other in 0.1s. Taking one end as x = 0 and the other end x = 2m and t = 0 as the time when P is at rest. (Position of P is x)
The ***CORRECT*** statements will be
(A) For $0 < t < 0.1$ s, energy flows across P in positive x-direction for $0 < x < 1$ m
(B) For $0 < t < 0.05$ s, energy flows across P in negative x-drection for $0 < x < 1$ m
(C) At t = 0.05 s, rate of energy flow through P is zero for x = 0.5 m
(D) At t = 0.1s, rate of energy flow through P for all values of x is zero

13. String of length L whose one end is at $x = 0$, vibrates according to the relations given by different equations. Choose the **CORRECT** statement(s).

(A) $y = A \sin\frac{\pi x}{L} \sin \omega t$ has 1 antinodes, 2 nodes

(B) $y = A \cos\frac{\pi x}{L} \sin \omega t$ has 2 antinodes, 1 nodes

(C) $y = A \sin\frac{2\pi x}{L} \sin \omega t$ has 3 nodes, 2 antinodes

(D) $y = A \cos\frac{2\pi x}{L} \sin \omega t$ has 3 antinodes, 2 nodes

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14. A wave disturbance in a medium is described by

$$y(x, t) = 0.02 \cos(50\pi t + \pi/2) \cos(10\pi x),$$

where x and y are in metres and t in seconds.

(A) A node occurs at x = 0.15 m
(B) An antinode occurs at x = 0.3 m
(C) The speed of the wave is 5.0 m/s
(D) The wavelength is 0.2 m

15. A waveform : $y_1 = A \sin\left(2x - 4t + \frac{\pi}{3}\right)$ is superposed with a second waveform, to produce a standing wave with a node at x = 0. The equation of the second waveform can be :-

(A) $y_2 = A \sin\left(2x + 4t + \frac{5\pi}{3}\right)$
(B) $y_2 = A \sin\left(2x - 4t + \frac{\pi}{3}\right)$
(C) $y_2 = A \sin\left(2x + 4t - \frac{\pi}{3}\right)$
(D) $y_2 = A \sin\left(2x + 4t + \frac{\pi}{3}\right)$

16. A string fixed at both ends and under tension T vibrates in its 1st overtone with an amplitude A at the antinodes. The total energy of the string is E and the maximum possible speed of a particle of the string is v. If the same string were to vibrate in its fundamental mode under a tension 4T and with an amplitude A at the antinode then :-

(A) The total energy of the string will be E
(B) The total energy of the string will be 2E
(C) The maximum possible speed of a particle on the string is v
(D) The maximum possible speed of a particle on the string is 2v

17. A long wire ABC is made by joining two wires AB and BC of equal cross-sectional area. AB has length 4.80 m and mass 0.12 kg. BC has length 2.56 m and mass 0.4 kg. The wire ABC is under a tension of 160 N. A sinusoidal wave y = 5.6 (cm) sin (ωt – kx) is sent along the wire ABC from the end A. No power dissipates during the propagation of the wave.



(A) The amplitude of the reflected wave is 2.4 cm
(B) The amplitude of transmitted wave is 3.2. cm
(C) The maximum displacement of the nodes of the stationary wave in the wire AB is 3.2 cm
(D) The fraction of power transmitted from the junction B is approximately 0.816

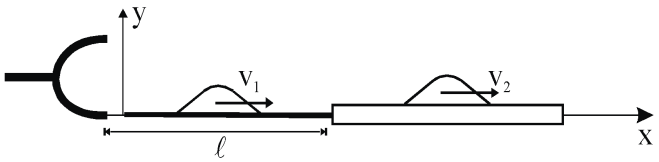
18. In a travelling one dimensional mechanical sinusoidal wave

(A) potential energy and kinetic energy of an element become maximum simultaneously.
(B) all particles oscillate with the same frequency and the same amplitude
(C) all particles may come to rest simultaneously
(D) we can find two particles, in a length equal to half of a wavelength, which have the same non zero acceleration simultaneously.

Linked Comprehension Type (2 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question Nos. 19 to 21

A harmonic oscillator at x = 0, oscillates with a frequency $\frac{\omega}{2\pi}$ and amplitude a. It is generating waves at end of a thin string in which velocity of wave is v_1 and which is connected to another heavier string in which velocity of wave is v_2 as shown, length of first string is ℓ.



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19. If harmonic oscillator oscillates by an equation y = a sinωt. The equation of incident wave in first string is

(A) $y = a\sin\omega\left(t-\frac{x}{v_1}\right)$ (B) $y = a\sin\omega\left(t+\frac{x}{v_1}\right)$

(C) $y=a\sin\left[\omega\left(t-\frac{x}{v_1}\right)+\pi\right]$ (D) $y=a\sin\left[\omega\left(t+\frac{x}{v_1}\right)+\pi\right]$

20. Equation of transmitted wave in second string if its amplitude is a_t is

(A) $y = a_t\sin\omega\left(t-\frac{x}{v_2}\right)$ (B) $y = a_t\sin\omega\left(t-\frac{\ell}{v_1}\right)$

(C) $y = a_t\sin\omega\left(t-\frac{\ell}{v_1}-\frac{x-\ell}{v_2}\right)$ (D) $y = a_t\sin\omega\left(t-\frac{x}{v_2}\right)$

21. Equation of reflected wave, if it is reflecting at the joint and amplitude of reflected wave is a_R

(A) $y = a_R\sin\omega\left(t-\frac{x}{v_2}\right)$ (B) $y=a_R\sin\left[\omega\left(t-\frac{\ell}{v_1}-\frac{\ell-x}{v_1}\right)+\pi\right]$

(C) $y=a_R\sin\left[\omega\left(t+\frac{x}{v_1}\right)+\pi\right]$ (D) $y=a_R\sin\left[\omega\left(t+\frac{2\ell+x}{v_1}\right)+\pi\right]$

Paragraph for Question Nos. 22 to 24

A 2m string has tension 1N is fixed at both end and its is vibrating in its third harmonic with antinode amplitude 3 cm and frequency 100 Hz, then

22. Possible stationary wave equation for the vibration of the string will be (assume origin at left end of the string and x is measured in meters towards right and t is measured in seconds)

(A) y = (0.06 m) $\sin\left(\frac{3\pi}{2}x\right)\cos(200\,\pi t)$ (B) y = (0.03 m) $\sin\left(\frac{3\pi}{4}x\right)\cos(200\,\pi t)$

(C) y = (0.06 m) $\sin\left(\frac{3\pi}{4}x\right)\cos(200\,\pi t)$ (D) y = (0.03 m) $\sin\left(\frac{3\pi}{2}x\right)\cos(200\,\pi t)$

23. Total wave energy on the string will be nearly equal to

(A) 40 mJ (B) 10 mJ (C) 30 mJ (D) 20 mJ

24. At what time from the start (by the answer of first question in this paragraph) string will have maximum kinetic energy first time (second)

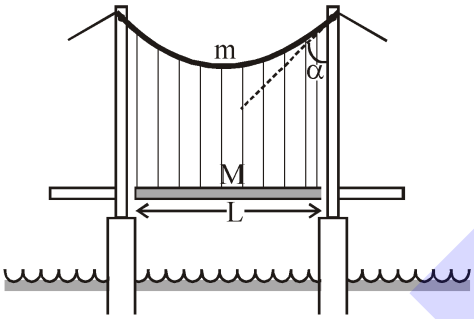
(A) $\frac{1}{200}$ (B) $\frac{1}{100}$ (C) $\frac{1}{400}$ (D) $\frac{1}{800}$

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Paragraph for Question No. 25 and 26

A suspension bridge consists of a pair of cables hung between two towers with the roadway suspended from these cables by means of closely spaced vertical wires of negligible mass as shown in figure. Each flexible cable of length ℓ (> L) and mass m hangs between two poles and mass of roadway section is M distributed uniformly along length L. At the ends, the cable makes an angle of α with the walls. (Assume each cable bears equal load and M>>m)



Hanging Bridge

25. The shape of any cable is best describe by equation assuming center of a cable is origin, vertical direction is taken as positive Y-axis and horizontal right taken as positive X-axis?

(A) $y = \frac{\cot\alpha}{2L}x^2$ (B) $y = \frac{\tan\alpha}{2L}x^2$ (C) $y = \frac{\tan\alpha}{L}x^2$ (D) $y = \frac{\cot\alpha}{L}x^2$

26. If a small amplitude transverse pulse is passed through the cable, what is the speed of transverse waves at the middle ?

(A) $\frac{1}{2}\sqrt{\frac{Mg\ell\tan\alpha}{m}}$ (B) $\sqrt{\frac{Mg\ell\tan\alpha}{m}}$ (C) $\frac{1}{2}\sqrt{\frac{Mg\ell\cot\alpha}{m}}$ (D) $\sqrt{\frac{Mg\ell\cot\alpha}{m}}$

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SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

27. A string will break apart if it is placed under too much tensile stress. One type of steel has density. $\rho_{steel} = 10^4$ kg/m^3 and breaking stress $\sigma = 9 \times 10^8$ N/m^2. We make a guitar string from (4π) gram of this type of steel. It should be able to withstand (900 π)N without breaking. What is highest possible fundamental frequency (in Hz) of standing waves on the string if the entire length of the string vibrates.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

28. A string of length ℓ is fixed at both ends. It is vibrating in its 3rd overtone with maximum amplitude $a = 2\sqrt{3}$ mm. Find the square of amplitude (in mm^2) at a distance $\frac{\ell}{3}$ from one end.

Subjective Type **1 Q. [4 M (0)]**

29. A steel of wire of length 25 cm is fixed at its ends to rigid walls. Young's modulus of steel = 200 GPa, coefficient of linear thermal expansion = 10^{-5} / °C. Initially, the wire is just taut at 20°C temperature. The density of steel = 8.0 g/cc. A tuning fork of frequency 200 Hz is touched to the wire, to execute oscillations. Simultaneously, the temperature is slowly lowered. At what temperature will resonance occur corresponding to the third overtone?

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GR # WAVE ON A STRING				ANSWER KEY
SECTION-I				
Single Correct Answer Type				**9 Q. [3 M (–1)]**
1. Ans. (C)	2. Ans. (C)	3. Ans. (D)	4. Ans. (B)	5. Ans. (C)
6. Ans. (A)	7. Ans. (B)	8. Ans. (C)	9. Ans. (B)	
Multiple Correct Answer Type				**9 Q. [4 M (–1)]**
10. Ans. (A,B,C,D)	11. Ans. (B, C, D)	12. Ans. (C,D)	13. Ans. (A,B,C,D)	14. Ans. (A, B, C, D)
15. Ans. (A, C)	16. Ans. (A, C)	17. Ans. (A,B,C,D)	18. Ans. (A,B,D)	
Linked Comprehension Type (Single Correct Answer Type)		**(2 Para × 3Q.)**	**(1 Para × 2Q.)**	**[3 M (-1)]**
19. Ans. (A)	20. Ans. (C)	21. Ans. (B)	22. Ans. (D)	23. Ans. (B)
24. Ans. (C)	25. Ans. (D)	26. Ans. (A)		
SECTION-II				
Numerical Answer Type Question (upto second decimal place)				**1Q.[3M (0)]**
27. Ans. 375				
SECTION-III				
Numerical Grid Type (Ranging from 0 to 9)				**1 Q. [4 M (0)]**
28. Ans. 9				
Subjective Type				**1 Q. [4 M (0)]**
29. Ans. 17.5°				

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